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6. Case Study: Food Delivery Application – Positive Testing

Aim: To identify and analyze the requirements of a Food Delivery Application and prepare positive test scenarios and test cases to verify its functionality.

Test Case ID	Test Scenario	Values Entered / Action Done	Expected Result	Actual Result	Status
TC01	User registration	Enter valid user details	Account created successfully	Account created successfully	Pass
TC02	User login	Enter valid credentials	Login successful	Dashboard displayed	Pass
TC03	Search restaurant	Search for “Pizza”	Matching restaurants displayed	Restaurants displayed	Pass
TC04	Select food item	Select an available food item	Food details displayed	Food details displayed	Pass
TC05	Add item to cart	Add a food item	Item added to cart	Item added successfully	Pass
TC06	Update quantity	Increase quantity to 2	Total price updated	Total price updated	Pass
TC07	Apply coupon	Enter a valid coupon code	Discount applied	Discount not applied	Fail
TC08	Place order	Confirm cart and delivery address	Order placed successfully	Order confirmation displayed	Pass
TC09	Online payment	Use valid payment details	Payment completed successfully	Payment failed unexpectedly	Fail
TC10	Track order	Open order tracking page	Current order status displayed	Order status displayed	Pass

Result: Positive test cases for the Food Delivery Application were prepared and executed using valid inputs. Most functions worked correctly, while coupon application and online payment revealed defects requiring correction.

7. Case Study: Food Delivery Application – Negative Testing

Aim: To identify and analyze the requirements of a Food Delivery Application and prepare negative test scenarios to verify its behavior under invalid inputs and unexpected conditions.

Test Case ID	Test Scenario	Values Entered / Action Done	Expected Result	Actual Result	Status
TC01	Invalid login	Enter an incorrect password	Error message displayed	Error message displayed	Pass
TC02	Empty login fields	Submit without credentials	Validation message displayed	Validation message displayed	Pass
TC03	Invalid restaurant search	Search for an unavailable restaurant	“No restaurants found” message displayed	No restaurants found	Pass
TC04	Unavailable food item	Add an out-of-stock item	Item rejected	Item added to cart	Fail
TC05	Invalid quantity	Enter zero or negative quantity	Invalid quantity rejected	Validation message displayed	Pass
TC06	Invalid coupon	Apply an expired coupon	Coupon rejected	Expired coupon accepted	Fail
TC07	Missing delivery address	Place an order without an address	Address required message displayed	Validation message displayed	Pass
TC08	Payment failure	Simulate a declined payment	Failure displayed and order handled safely	Failure message displayed	Pass
TC09	Duplicate order	Submit the same order repeatedly	Duplicate order prevented	Duplicate order created	Fail
TC10	Network failure	Disconnect during checkout	Error displayed and transaction handled safely	Error displayed; order status remained unclear	Fail

Result: Negative testing identified defects in stock validation, coupon validation, duplicate order prevention, and network failure handling. These issues must be fixed and retested before deployment.

8. Case Study: Banking Application – Positive Testing

Aim: To identify and analyze the requirements of a Banking Application and prepare positive test scenarios and test cases to verify its major banking operations.

Test Case ID	Test Scenario	Values Entered / Action Done	Expected Result	Actual Result	Status
TC01	Valid login	Enter correct customer ID and password	Login successful	Dashboard displayed	Pass
TC02	Balance enquiry	Select balance enquiry	Correct balance displayed	Balance displayed	Pass
TC03	Fund transfer	Transfer ₹1,000 to a valid beneficiary	Transfer successful	Transfer confirmation displayed	Pass
TC04	Cash withdrawal	Withdraw ₹2,000 with sufficient balance	Withdrawal completed	Transaction completed	Pass
TC05	Cash deposit	Deposit ₹5,000	Balance updated	Balance updated	Pass
TC06	Transaction history	Open transaction history	Previous transactions displayed accurately	Transactions displayed	Pass
TC07	Add beneficiary	Enter valid beneficiary details	Beneficiary added successfully	Beneficiary not added	Fail
TC08	Bill payment	Pay a valid electricity bill	Payment successful	Receipt displayed	Pass
TC09	Change password	Enter valid current and new passwords	Password updated	Password update failed	Fail
TC10	Logout	Select logout	Session terminated	Login page displayed	Pass

Result: Positive test cases were prepared to verify banking operations using valid inputs. Defects were identified in beneficiary registration and password updating, requiring further investigation and correction.

9. Case Study: Banking Application – Negative Testing

Aim: To identify and analyze the requirements of a Banking Application and prepare negative test scenarios to verify its handling of invalid transactions, unauthorized access, and exceptional conditions.

Test Case ID	Test Scenario	Values Entered / Action Done	Expected Result	Actual Result	Status
TC01	Invalid login	Enter incorrect password	Access denied	Error message displayed	Pass
TC02	Empty credentials	Submit without a password	Validation message displayed	Validation message displayed	Pass
TC03	Repeated incorrect login	Enter incorrect credentials repeatedly	Login restricted according to security policy	Access remained available after repeated failures	Fail
TC04	Insufficient balance	Transfer ₹10,000 with ₹5,000 balance	Transfer rejected without debit	Transfer rejected	Pass
TC05	Invalid transfer amount	Enter ₹-500	Invalid amount rejected	Validation message displayed	Pass
TC06	Invalid beneficiary	Transfer to an invalid account	Transfer rejected	Transfer rejected	Pass
TC07	Exceed transaction limit	Transfer above the permitted limit	Transaction rejected or additional verification required	Transfer processed without required verification	Fail
TC08	Duplicate transaction	Submit the same transfer twice	Duplicate debit prevented	Duplicate debit occurred	Fail
TC09	Unauthorized access	Attempt to access another customer's account	Access denied	Access denied	Pass
TC10	Network failure	Disconnect during a fund transfer	Transaction handled safely without inconsistent balance	Transfer status remained unclear	Fail

Result: Negative testing revealed security and transaction-handling defects involving repeated login failures, transaction limits, duplicate transfers, and network interruptions. These defects require urgent correction because they may affect account security and financial integrity.

10. Comprehensive Test Plan: Library Management System

Aim: To develop a comprehensive Test Plan for a Library Management System, covering testing strategies, methodologies, and test cases to evaluate its functionality, security, and reliability.

Test Case ID	Test Scenario	Values Entered / Action Done	Expected Result	Actual Result	Status
TC01	Valid login	Enter valid librarian credentials	Dashboard displayed	Dashboard displayed	Pass
TC02	Invalid login	Enter incorrect password	Access denied	Error message displayed	Pass
TC03	Add a book	Enter valid book details	Book added successfully	Book added successfully	Pass
TC04	Search a book	Enter valid title or ISBN	Matching book displayed	Book displayed	Pass
TC05	Register member	Enter valid member details	Member registered successfully	Member registration failed	Fail
TC06	Issue an available book	Select eligible member and available book	Book issued and availability updated	Book issued successfully	Pass
TC07	Issue unavailable book	Select a book with no copies available	Issue rejected or waitlist offered	Unavailable book was issued	Fail
TC08	Return a book	Return an issued book	Loan closed and inventory updated	Loan closed successfully	Pass
TC09	Calculate overdue fine	Return a book after its due date	Fine calculated according to configured rules	Fine calculated incorrectly	Fail
TC10	Renew a book	Renew an eligible loan	Due date updated	Due date updated	Pass
TC11	Duplicate book record	Enter a duplicate ISBN where duplicates are prohibited	Duplicate rejected	Duplicate book record created	Fail
TC12	Unauthorized deletion	Member attempts to delete a book	Access denied	Access denied	Pass
TC13	Invalid member ID	Issue a book to a nonexistent member	Operation rejected	Operation rejected	Pass
TC14	Empty mandatory fields	Submit a blank book form	Validation messages displayed	Validation messages displayed	Pass

TC15	Generate overdue report	Request an overdue-book report	Accurate overdue records displayed	Report omitted some overdue books	Fail

Result: The Library Management System test plan was prepared with functional test cases covering authentication, book management, member registration, issue and return operations, fine calculation, and report generation. Testing identified defects in member registration, inventory validation, fine calculation, duplicate book prevention, and overdue reports. These defects should be corrected and the affected test cases rerun before the system is approved for release.